

# 220: Midterm 1 Study Guide

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## Facts

### T1: Topological Spaces

**Example.** • If a topological space  $\mathcal{T}$  is **metrizable**, then the metric that induces  $\mathcal{T}$  is not unique.

- The coarsest topology is the **indiscrete topology** and the finest topology is the power set of a given nonempty set.
- For any topology  $\mathcal{T}$ , we have the following inclusions:

$$\{\emptyset, X\} \subseteq \mathcal{T} \subseteq \mathcal{P}(X)$$

- $\mathcal{T} = \{(a, b) : a, b \in \mathbb{R}, a < b\} \cup \{\emptyset, \mathbb{R}\}$  is not a topology on  $\mathbb{R}$  because  $(0, 1)$  and  $(2, 3)$  are in  $\mathcal{T}$  but their union  $(0, 1) \cup (2, 3)$  is not in the form  $(a, b)$  and so  $(0, 1) \cup (2, 3) \notin \mathcal{T}$ .

### T2: Bases For Topological Spaces

**Example.** • If  $(X, \mathcal{T})$  is a topological space, then  $\mathcal{T}$  is basis for itself.

- In a metric space, the collection of all open balls is a basis for the topology induced by the metric.
- If  $X$  is nonempty, the collection of all singletons is a basis for the discrete topology on  $X$ .
- The open balls generated by  $d_2$  and  $d_\infty$  metric form a basis for the Euclidean topology on  $\mathbb{R}^2$ .
- Consider the following bases

$$\mathcal{B}_E = \{(a, b) : a < b\}, \quad \mathcal{B}_L = \{[a, b) : a < b\}, \quad \mathcal{B}_K = \mathcal{B} \cup \{(a, b) \setminus K : a < b\}$$

where  $K = \{1/n : n \in \mathbb{N}\}$ . These bases generate the Euclidean topology  $\mathcal{T}_E$ , the Lower Limit Topology  $\mathcal{T}_L$ , and the  $K$ -topology  $\mathcal{T}_K$ .

The lower limit topology and the  $K$ -topology are strictly finer than the Euclidean one, but are not comparable to each other.

**Remark.** Let  $X$  be a nonempty set. If a collection of subsets  $\mathcal{B}$  of  $X$  satisfies the following properties:

- (i)  $\bigcup_{B \in \mathcal{B}} B = X$
- (ii) For all  $A, B \in \mathcal{B}$ , then  $A \cap B$  can be written as a union of elements in  $\mathcal{B}$

Then there exists a unique topology  $\mathcal{T}$  such that  $\mathcal{B}$  generates  $\mathcal{T}$ .

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**Remark.** In general, we can determine whether a collection of subsets of  $X$  forms a basis for the topology. However, if the collection happens to satisfy the above properties, then it is a basis for some topology that is unique.

### T3: Neighborhoods, Interior

**Example** (Neighborhoods That Are Open/Not Open). • For all  $r > 0$ ,  $(x - r, x + r)$  is an open neighborhood containing  $x$  since it is an open set containing  $x$ . On the other hand,  $[x - r, x + r)$  is not open and contains the open set  $(x - r, x + r)$ . Thus,  $[x - r, x + r)$  is a neighborhood but is not open.

- For all  $r > 0$ ,  $[x, x + r)$  is not a neighborhood of  $x$ , otherwise there would be an open interval  $(x - t, x + t) \subseteq [x, x + r)$  for some  $t > 0$ , which is impossible. We can see this by looking at the left-endpoint; that is, any open set around  $x$  will not be contained within  $[x, x + r)$ .

**Example** (Computing Interiors of Sets in Different Topologies). Given  $A = (a, b)$ ,  $B = [a, b)$ ,  $C = [a, b]$ , and  $D = \mathbb{Q}$ . Find the interior of  $A, B, C, D$  as subsets of  $\mathbb{R}$  endowed with

- **The Euclidean Topology:**  $\overset{\circ}{A} = A$ ,  $\overset{\circ}{B} = A$ ,  $\overset{\circ}{C} = A$ , and  $\overset{\circ}{D} = \emptyset$ ;
- **The Lower Limit Topology:**  $\overset{\circ}{A} = A$ ,  $\overset{\circ}{B} = B$ ,  $\overset{\circ}{C} = B$ , and  $\overset{\circ}{D} = \emptyset$ ;
- **The Cofinite Topology**  $\overset{\circ}{A} = \overset{\circ}{B} = \overset{\circ}{C} = \overset{\circ}{D} = \emptyset$

### T4: Closed Sets and Closure

**Example** (Closed Sets in  $\mathbb{R}$ ). The sets  $[a, \infty)$ ,  $(-\infty, a]$ ,  $[a, b]$  for  $a, b \in \mathbb{R}$  with  $a < b$  are closed in  $\mathbb{R}$  equipped with the Euclidean Topology.

It suffices to observe that

- $\mathbb{R} \setminus [a, \infty) = (-\infty, a)$  which is clearly open in  $\mathbb{R}$  and so we see that  $[a, \infty)$  is closed in  $\mathbb{R}$ .
- $\mathbb{R} \setminus (-\infty, a] = (a, \infty)$  which is clearly open in  $\mathbb{R}$  and so we see that  $(-\infty, a]$  is closed in  $\mathbb{R}$ .
- $\mathbb{R} \setminus [a, b] = (-\infty, a) \cup (b, \infty)$  is a union of open sets in  $\mathbb{R}$  and so the complement of  $[a, b]$  is open in  $\mathbb{R}$ ; that is,  $[a, b]$  is a closed set in  $\mathbb{R}$ .

**Example** (Non-examples). The sets  $[a, b)$ ,  $(a, b]$  for  $a, b \in \mathbb{R}$  with  $a < b$  are not closed in  $\mathbb{R}$  equipped with the Euclidean Topology.

- Note that  $\mathbb{R} \setminus [a, b) = (-\infty, a) \cup [b, \infty)$  is not open because  $[b, \infty)$  is not an open set in  $\mathbb{R}$ . Indeed, we can find an open ball centered at  $b$  such that it is not completely contained in  $[b, \infty)$ . That is,  $b$  is not an interior point of the complement. Hence, we conclude that  $[a, b)$  is not a closed set in  $\mathbb{R}$ .
- Likewise,  $\mathbb{R} \setminus (a, b] = (-\infty, a] \cup (b, \infty)$  since  $a$  is not an interior point of  $\mathbb{R} \setminus (a, b]$ .

**Example.** In a metric space, singletons are closed.

**Example.** Which of the sets in the previous example are closed in  $\mathbb{R}$  equipped with the lower limit topology?

- $\mathbb{R} \setminus [a, +\infty) = (-\infty, a) = \bigcup_{n \in \mathbb{N}} [a - n, a)$  is open since we can write the complement of  $[a, \infty)$  as a union of basis elements of  $\mathcal{T}_L$ . That is,  $[a, +\infty)$  is closed in the lower limit topology.
- $\mathbb{R} \setminus (-\infty, a] = (a, +\infty) = \bigcup_{n \in \mathbb{N}} [a + \frac{1}{n}, a + n)$  is open in  $\mathbb{R}$  and so  $[a, +\infty)$ .
- Similarly,  $\mathbb{R} \setminus [a, b) = (-\infty, a) \cup [b, +\infty)$  is open as a union of open sets, so  $[a, b)$  is closed in the Lower Limit Topology.
- $\mathbb{R} \setminus (a, b] = (-\infty, a] \cup (b, +\infty)$  is not open since  $a$  is not an interior point of the complement; that is, there does not exist an interval of the form  $[c, d)$  with  $c, d \in \mathbb{R}$  with  $c < d$ ,  $a \in [c, d)$  that is contained in  $A$ . Thus, it follows that  $(a, b]$  is not closed.

**Example.** Which of the following sets are closed in  $\mathbb{R}^2$  equipped with the Euclidean topology?

- (i)  $C = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 \leq 1\}$ ;

Notice that  $\mathbb{R}^2 \setminus C = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 > 1\}$ . We can prove that this set is open in  $\mathbb{R}^2$  using the same methods found in homework 1. Hence, it follows that  $C$  is closed in  $\mathbb{R}^2$ .

- (ii)  $A = \{0\} \times \mathbb{R}$ .

Again, we can show that  $\mathbb{R}^2 \setminus A$  is open.

**Example (Dense Sets In A Topological Space).** (i) The set of rationals  $\mathbb{Q}$  is dense in  $\mathbb{R}$  equipped with the Euclidean Topology (and with the Lower Limit Topology)

- (ii) More in general, for  $n \in \mathbb{N}$ , the set  $\mathbb{Q}^n$  is dense in  $\mathbb{R}^n$  equipped with the Euclidean Topology.

- (iii) The set of irrationals  $\mathbb{R} \setminus \mathbb{Q}$  is dense in  $\mathbb{R}$  with the Euclidean Topology.

## T5: Limit and Boundary Points

**Example.** In  $\mathbb{R}$  equipped with the Euclidean topology find the derived of  $A = (-\infty, 1) \cup \{2\}$  and  $B = \{n^{-1} : n \in \mathbb{N}\}$ .

**Example.** In  $\mathbb{R}$  equipped with the Euclidean topology find the boundary of the following sets

- (1)  $A = [a, b]$  where  $a, b \in \mathbb{R}$  with  $a < b$ ;
- (2)  $B = (a, b)$  where  $a, b \in \mathbb{R}$  with  $a < b$ ;
- (3)  $C = \mathbb{Q}$ .

**Example.** Let  $(X, d)$  be a metric space  $x \in X$ ,  $r > 0$ . We have that

$$\partial B_r(x) \subseteq \{y \in X : d(y, x) = r\}.$$

Does equality necessarily hold?

**Example (Finding the Interior, Closure, Derived, and Boundary).** (1)  $A = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 \neq 0\}$

- (2)  $B = \left\{ (x_1, x_2) \in \mathbb{R}^2 : \frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} < 1 \right\}$  where  $a, b \in \mathbb{R}$ .

$$(3) \ C = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 = \sin(x_1^{-1}), x_1 > 0\}.$$

$$(4) \ D = \mathbb{Q} \times \mathbb{R}$$

## Definitions

**Definition (Metric).** Let  $X$  be a nonempty set. A **metric** (or **distance**) on  $X$  is a function  $d : X \times X \rightarrow [0, \infty)$  such that for all  $x, y, z \in X$

$$(i) \ d(x, y) = 0 \text{ if and only if } x = y$$

$$(ii) \ d(x, y) = d(y, x)$$

$$(iii) \ d(x, y) \leq d(x, z) + d(z, y)$$

The pair  $(X, d)$  is called a **metric space**.

**Definition (Open Ball in a Metric Space).** Let  $(X, d)$  be a metric space,  $x_0 \in X$ . We define the **open ball** centered  $x_0$  with radius  $r > 0$  as

$$B_r(x_0) = \{x \in X : d(x, x_0) < r\}.$$

**Definition (Open Set in a Metric Space).** Let  $(X, d)$  be a metric space. We say that a nonempty set  $A \subseteq X$  is open if for all  $x \in A$ , there exists a  $\delta > 0$  such that  $B_\delta(x) \subseteq A$ .

**Definition (Continuity In A Metric Space).** Let  $(X, d_X)$  and  $(Y, d_Y)$  be two metric spaces. We say that a function  $f : X \rightarrow Y$  is **continuous** at  $x_0 \in X$  if for all  $\varepsilon > 0$ , there exists a  $\delta > 0$  such that if  $d_X(x, x_0) < \delta$ , then  $d_Y(f(x), f(x_0)) < \varepsilon$ .

We say that  $f$  is continuous on  $X$  if  $f$  is continuous for all  $x_0 \in X$ .

**Definition (Topology).** Let  $X$  be a nonempty set. A **topology** on  $X$  is a collection of subsets of  $X$  denoted by  $\mathcal{T}$  such that

$$(i) \ \emptyset, X \in \mathcal{T};$$

$$(ii) \ \text{If } A_1, A_2 \in \mathcal{T}, \text{ then } A_1 \cap A_2 \in \mathcal{T};$$

$$(iii) \ \text{If } \{A_i\}_{i \in I}, \text{ then } \bigcup_{i \in I} A_i \in \mathcal{T}.$$

**Definition (Metrizable Topological Spaces).** We say that a topological space  $(X, \mathcal{T})$  is **metrizable** if there exists a metric  $d$  that induces the topology  $\mathcal{T}$ .

**Definition (Basis of a Topology).** Let  $(X, \mathcal{T})$  be a topological space. A **basis** for  $\mathcal{T}$  is a collection of open sets  $\mathcal{B} \subseteq \mathcal{T}$  such that any set  $U \in \mathcal{T}$  can be written as a union of sets in  $\mathcal{B}$ ; that is, for all  $U \in \mathcal{T}$ ,

$$U = \bigcup_{B \in \mathcal{B}} B.$$

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**Definition** (Neighborhood). Let  $(X, \mathcal{T})$  be a topological space and  $x \in X$ . A **Neighborhood** of  $x$  is a subset  $N$  of  $X$  such that there exists  $U \in \mathcal{T}$  such that  $x \in U \subseteq N$ . The family of all neighborhoods of  $x$  is called the **neighborhood system** of  $x$  and is denoted by  $\mathcal{N}(x)$ .

**Definition** (Interior). Let  $X$  be a topological space,  $A \subseteq X$ , and  $x \in X$ . We say that  $x$  is an **interior point** of  $A$  if there exists an open set  $U \subseteq X$  containing  $x$  such that  $x \in U \subseteq A$ .

The **interior** of  $A$  is the set of all interior points of  $A$  denoted by

$$\text{int}(A) = \{x \in A : \exists U \subseteq X \text{ such that } x \in U \subseteq A\}.$$